

S5

Hypothesis Testing

Learning Objectives

In this chapter, we will

- state the definitions of null hypothesis (H_0) and alternative hypothesis (H_1), test statistic, critical (rejection) region, critical value, level of significance and p -value;
- state the test statistic for a hypothesis test and define the necessary parameters;
- formulate hypotheses and test for a population mean based on
 - a (random) sample taken from a normal population of known variance
 - a large (random) sample taken from any population (including normal distribution) with the use of the Central Limit Theorem
- distinguish between 1-tail and 2-tail tests;
- interpret the results of a hypothesis test in the context of the problem.

5.1 Introduction - Hypothesis Testing: Making Decisions with Data

Statistical inference generally has two major branches:

- Estimation (Chapter S4)** – using a **random sample** data to **estimate** population parameters (e.g., estimating population mean, μ , and population variance, σ^2).
- Hypothesis testing** – using a **random sample** data to **test claims** about population parameters e.g. $X \sim N(\mu, \sigma^2)$. In our syllabus, we focus specifically on testing a population mean where our **test statistic** is our sample mean, $\bar{X}$.

5.1.1 Setting Up the Hypotheses

In every hypothesis test, we must choose between **two** opposing statements: the null hypothesis (H_0) and the alternative hypothesis (H_1).

Let us consider the following example to illustrate the idea of a hypothesis test.

The **label on a bottle** of X-Cola states that the **volume** is 500 ml. Due to a technical issue in the production line, the **manager suspects that the mean volume** of X-Cola bottles **is less than** 500 ml. To investigate his suspicion, the manager collects a **random sample** of 50 bottles

and measures their contents. The sample has a mean volume of 499.3 ml.

Question: Does this **sample mean (test statistic value)** of 499.3 ml provide enough (significant) evidence to conclude that the actual (population) mean volume of all bottles of X-Cola is less than 500ml?

Two Possible Conclusions:

- (a) The **sample mean (test statistic value)** of 499.3 ml is close to 500 ml. The lower value could reasonably be explained by normal statistical variation. If every bottle were measured, we would not be surprised if the population mean were indeed 500 ml.
- (b) The **sample mean (test statistic value)** of 499.3 ml is sufficiently far from 500 ml that the difference is unlikely to have occurred due to statistical variation alone. If every bottle were measured, we would be surprised if the population mean were still 500 ml. In this case, we have evidence to suggest that the actual population mean is less than 500 ml.

So, how would you decide if 499.3ml is close or far from 500ml?

In this chapter, we will learn how to make such decisions by conducting a full hypothesis test supported by statistical theory.

5.2 Key Concepts in Hypothesis Testing

After understanding the motivation behind hypothesis testing, we now define some key ideas formally.

Let us **assume** that X , the volume of drink in a bottle of X-Cola in ml, follows a **normal distribution** with **population standard deviation** 2. i.e. $X \sim N(500, 2^2)$

5.2.1 Null Hypothesis (H_0) vs Alternative Hypothesis (H_1)

The **null hypothesis** is the statement we assume to be true unless there is strong evidence against it. It states that the **population mean** μ is **equal** to a specified assumed population mean μ_0 .

- Example: The label on a bottle of X-Cola states that the volume is 500 ml, i.e. $H_0 : \mu = 500$.

The **alternative hypothesis** is the statement we are trying to gather evidence to support. If we have sufficient evidence, we will reject H_0 in favor of H_1 .

There are **three possible ways to formulate** the alternative hypothesis, H_1 , depending on what we suspect is happening to the **population mean** μ as shown in Table 1 below.

Whether the test is one-tailed or two-tailed depends on the form of the alternative hypothesis (H_1).

To test Null Hypothesis (H_0)	Against Alternative Hypothesis (H_1)
E.g. The label on a bottle of X-Cola states that the volume is 500 ml. $H_0 : \mu = 500$	E.g. The manager suspects the population mean volume of X-Cola is less than 500ml $H_1 : \mu < 500$ Left-tailed test (one-tailed)
	E.g. The manager suspects the population mean volume of X-Cola is more than 500ml $H_1 : \mu > 500$ Right-tailed test (one-tailed)
	E.g. The manager suspects the population mean volume of X-Cola is not 500ml $H_1 : \mu \neq 500$ Two-tailed test

Table 1

5.2.2 Test Statistic

X , the volume of drink in a bottle of X Cola in ml, follows a **normal distribution** with **population standard deviation 2**. i.e. $X \sim N(500, 2^2)$

A **test statistic** is a **random variable** (i.e. $\bar{X}$ or Z) calculated from sample data and used to decide whether to reject the null hypothesis.

In this case, test Statistic, $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$.

A **test statistic value** is a **specific reading of the test statistic** (i.e. $\bar{x}$ or z) calculated from a known sample.

In this case, the sample has a mean volume of 499.3 ml. **Test Statistic value** $\bar{x} = 499.3$

5.2.3 Level of Significance

A **Type I error** occurs in hypothesis testing when the Null Hypothesis H_0 is wrongly rejected when it is actually true. This wrong decision happens if the **sample mean (test statistic) value** falls within the **critical (rejection) region**.

Thus, level of significance, $\alpha\% = P(\text{rejecting } H_0 \mid H_0 \text{ is true})$.

Let us set our level of significance at 5% for our X-Cola example.

The level of significance $\alpha\%$ is a cut-off point chosen **before** collecting sample data.

- It represents the probability of wrongly rejecting the null hypothesis H_0 when it is actually true. Hence the value of $\alpha\%$ chosen is normally small (e.g. $\alpha\% = 10\%$).

In the context of the X-Cola example, 5% level of significance means the probability of **wrongly concluding** that the **population mean volume** of X-Cola is **less than** 500 ml when it is in fact 500 ml, is 0.05.

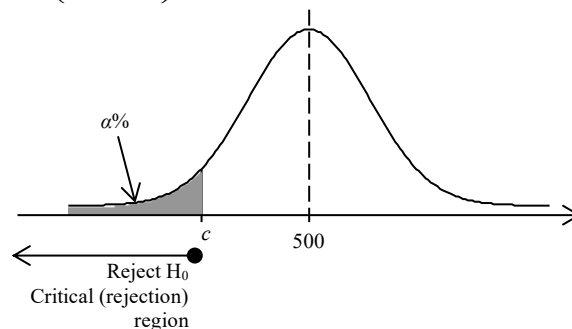
5.2.4 Critical (Rejection) Region and Critical Value

Critical Value c or z_α : This is the **boundary point on the test statistic's sampling distribution** $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ or standardised to $Z \sim N(0,1)$.

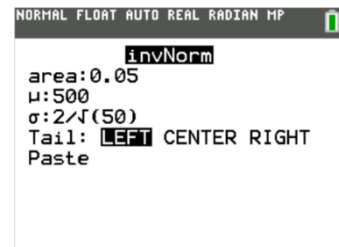
The critical value separates the rejection region from the rest of the distribution.

In the X-Cola example, under H_0 , i.e. assuming $H_0: \mu = 500$ is true, we have

Test Statistic, $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$.



We may calculate the **critical value c** by using GC as shown on the right, and we will get $c = 499.53$.

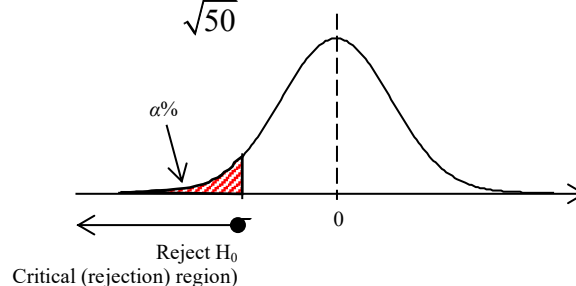


Since our collected **sample mean (test statistic value)** $\bar{x} = 499.3 < c$, $\bar{x}$ falls in the critical (rejection) region and we reject H_0 .

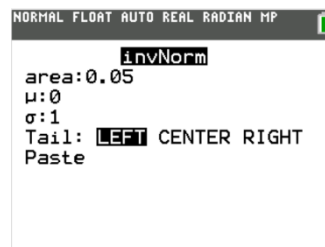
We conclude that $\mu < 500$ at 5% level of significance.

Instead of working with the sample mean $\bar{X}$ distribution, we may also choose to standardize $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$ to $Z \sim N(0,1)$, which is more commonly done and necessary in 'reverse testing' (when we are given the conclusion and asked to find a value instead).

Alternatively, Test Statistic, $Z = \frac{\bar{X} - 500}{\frac{2}{\sqrt{50}}} \sim N(0,1)$



We may calculate the **critical value** z_α by using GC as shown on the right, and we will get $z_\alpha = -1.6449$.



Since our collected **sample mean (test statistic value)**

$$z = \frac{499.3 - 500}{\frac{2}{\sqrt{50}}} = -2.4749 < z_\alpha, \quad z \text{ falls in the critical (rejection)}$$

region and we reject H_0 .

We conclude that $\mu < 500$ at 5% level of significance.

In Summary,

Left-tailed Test $H_0 : \mu = 500$ $H_1 : \mu < 500$	Right-tailed Test $H_0 : \mu = 500$ $H_1 : \mu > 500$	Two-tailed Test $H_0 : \mu = 500$ $H_1 : \mu \neq 500$
Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical value , $c = 499.53$. Using $Z \sim N(0,1)$, critical value , $z_\alpha = -1.64$.	Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical value , $c = 500.47$. Using $Z \sim N(0,1)$, critical value , $z_\alpha = 1.64$.	Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical values , $c = 499.45, 500.55$. Using $Z \sim N(0,1)$, critical values , $z_\alpha = \pm 1.96$.
Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical value , $\bar{x} \leq 499.53$. Using $Z \sim N(0,1)$, critical (rejection) region , $z \leq -1.64$.	Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical (rejection) region , $\bar{x} \geq 500.47$. Using $Z \sim N(0,1)$, critical (rejection) region , $z \geq -1.64$.	Using $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$, critical (rejection) region , $\bar{x} \leq 499.45$ or $\bar{x} \geq 500.55$. Using $Z \sim N(0,1)$, critical (rejection) regions , $z \leq -1.96$ or $z \geq -1.96$.

5.2.5 p -value

Instead of comparing the **sample mean (test statistic value)** with the **critical value** (e.g. c and z_α) we can **compare probabilities**.

The **p -value** is the probability, assuming the null hypothesis (H_0) is true, of getting a test statistic value **as extreme as or more extreme** than the one obtained from the sample data.

p -value

= **P(obtaining a test statistic as extreme or more extreme than the observed value | H_0 is true)**

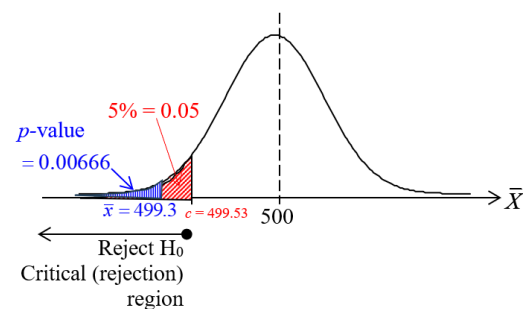
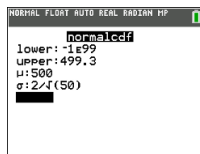
In the X-Cola example, if the manager suspects that the mean volume of X-Cola bottles is less than 500 ml, we will carry a left-tailed test to test the hypothesis.

The significance level is 5%. Under H_0 , i.e. assuming $H_0: \mu = 500$ is true, we have

Test Statistic, $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$.

The sample has a mean volume of 499.3 ml. **Test Statistic value** $\bar{x} = 499.3$

$$p\text{-value} = P(\bar{X} < \bar{x}) = P(\bar{X} < 499.3) = 0.00666.$$



Alternatively, Test Statistic, $Z = \frac{\bar{X} - 500}{\frac{2}{\sqrt{50}}} \sim N(0,1)$

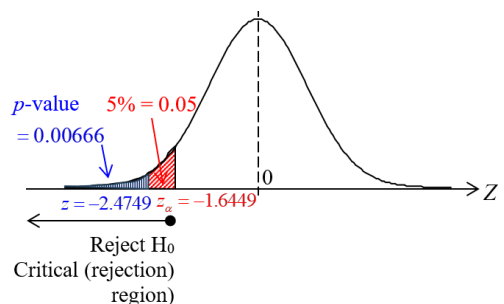
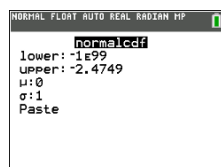
p -value

$$= P(Z < z)$$

$$= P\left(Z < \frac{499.3 - 500}{\frac{2}{\sqrt{50}}}\right)$$

$$= P(Z < -2.4749)$$

$$= 0.00666$$



Since $p\text{-value} = 0.00666 < 0.05 = \frac{\alpha}{100}$, we reject H_0 .

We have sufficient evidence to conclude that $\mu < 500$ at 5% level of significance.

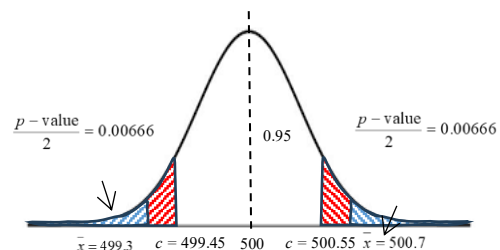
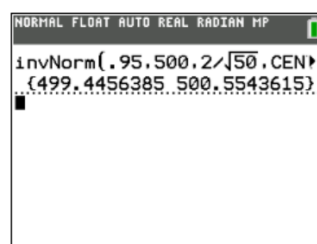
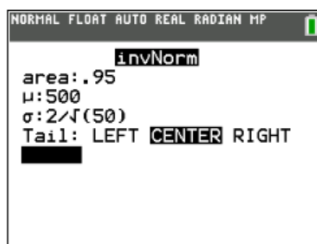
In the context of the X-Cola example, the **p -value** of 0.00666 represents the probability of obtaining a **sample mean volume** of **499.3 ml or less**, when the **population mean volume** is in fact **500 ml**.

In the X-Cola example, if the manager suspects that the mean volume of X-Cola bottles is not 500 ml, we will carry a 2-tailed test to test the hypothesis. The **significance level** 5% is therefore **split equally between both tails**, forming **two critical (rejection) regions**.

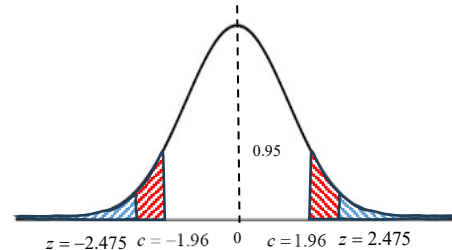
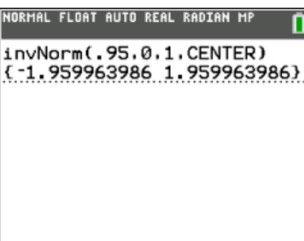
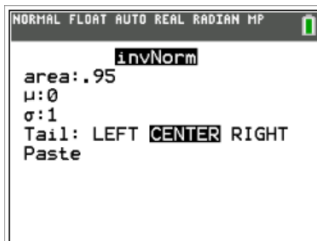
Under H_0 , i.e. assuming $H_0: \mu = 500$ is true, we have **Test Statistic**, $\bar{X} \sim N\left(500, \frac{2^2}{50}\right)$.

The sample has a mean volume of 499.3 ml. **Test Statistic value** $\bar{x} = 499.3$

There will be **two critical values**, 499.45 and 500.55, due to the **symmetry** of the standard normal distribution about mean 500.



Alternatively, Test Statistic, $Z = \frac{\bar{X} - 500}{\frac{2}{\sqrt{50}}} \sim N(0,1)$



$p\text{-value} = 2 \times P(\bar{X} < \bar{x}) = 2 \times P(\bar{X} < 499.3) = 2 \times 0.00666 = 0.01332 < 0.05$, we reject H_0 .

In the context of the X-Cola example, the **p-value** of 0.01332 represents the probability of obtaining a **sample mean volume less than or equal to 499.3 ml, or greater than or equal to 500.7ml**, when the **population mean volume** is in fact **500 ml**.

🔍 Note

1. A smaller p -value provides stronger evidence against H_0 .
2. Reject H_0 if $p\text{-value} \leq \frac{\alpha}{100}$; **do not reject** H_0 if $p\text{-value} > \frac{\alpha}{100}$.
3. The p -value shows the **smallest level of significance** at which H_0 would be rejected.

In general:

For a **one-tailed test**, the direction of the inequality used to calculate the p -value **follows the direction of the inequality of the alternative hypothesis H_1** .

For a **two-tailed test**, we are interested in whether the test statistic lies **far from the mean in either direction**.

- The **significance level $\alpha\%$** is therefore **split equally between both tails**, forming **two critical (rejection) regions**.
- There will be **two critical values**, $z_{\alpha/2}$ and $-z_{\alpha/2}$, due to the **symmetry** of the standard normal distribution about mean 0.
- If the test statistic falls in the **lower tail**, then the **p -value** is given by $p\text{-value} = 2 \times P(Z < z)$, as shown in Figure 6 below.

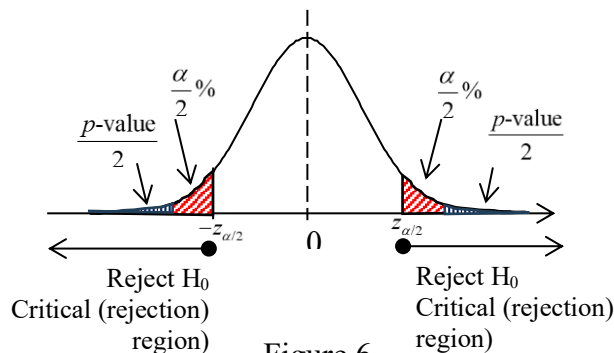


Figure 6

Note

1. It is helpful to **draw the relevant bell curve** for every hypothesis test question. This strengthens your **conceptual understanding** of the topic.
2. There are **different valid approaches** to hypothesis testing. To obtain **full marks** in exam questions, it is recommended to **follow the standard presentation format** shown in the upcoming examples.

Example 1

In the following situations, state suitable null and alternative hypotheses.

- (a) The mean factory assembly time for a particular electronic component is 84 seconds. The manager wants to test whether the new process results in a different assembly time.
- (b) The reported average age of hospital patients is 53 years. The director wants to test whether this figure is an **underestimate**.

Solution

- (a) Let μ be the **population mean** factory assembly time (in seconds).

To test $H_0: \mu = 84$ against $H_1: \mu \neq 84$

- (b) Let μ be the **population average** age (in years) of the hospital patients.

To test $H_0: \mu = 53$ against $H_1: \mu > 53$

🔍 Note

1. Example 1(a) is a **two-tailed test** because H_1 proposes a difference in **either direction** from the stated population mean.
2. Example 1(b) and the **X-Cola example in 5.1** are **one-tailed tests** because their alternative hypotheses propose a difference in **only one direction** (either greater than or less than the stated population mean).

5.3 # Z-Test and Examples

Since our test statistic is **Z** after standardizing the random variable $\bar{X}$, this type of hypothesis test is called a **z-test**. Let us consider the following cases.

(A): Sample from a population following a Normally distribution with known population variance

- Applicable when the population is normally distributed with known population variance σ^2 , i.e., $X \sim N(\mu, \sigma^2)$.
- In this case, under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$, the test statistic is calculated as

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Example 2

A machine fills sand into bags. The mass of a filled bag is normally distributed with a mean of 1500 g and a variance of 0.16 g². A random sample of 10 bags has a mean mass of 1499.8 g.

- (a) Explain what is meant by the term “random sample” in the context of this question and explain why it is necessary.
- (b) At the 1% significance level, test whether the sample provides enough evidence to conclude that the machine **underfills the bags**.
- (c) Find the **smallest level of significance (to 1 decimal place)** at which the sample would lead to the conclusion that the machine **underfills the bags**.

Solution

- (a) Every filled bag has an equal chance of being selected, and every filled bag is selected independently of the filled bags. A random sample therefore is an unbiased sample that is fair for us to use to infer about the population.

(b)

Step 1: Define the random variable and symbols

Let X be the mass (in grams) of a filled bag.

Let μ and σ^2 be the population mean and variance of X .

Step 2: State the data given

$$\bar{x} = 1499.8, \mu = 1500, \sigma^2 = 0.16, n = 10$$

Step 3: State the null and alternative hypotheses

To test $H_0 : \mu = 1500$ against $H_1 : \mu < 1500$

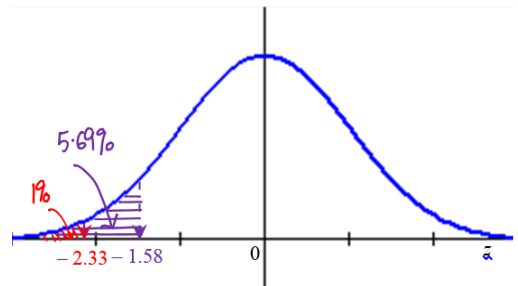
Note: this is a left-tailed test

Step 4: State the distribution of sample mean $\bar{X}$

Under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Step 5: Define the test statistic

$$\text{Test statistic, } Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

**Step 6: State the level of significance**

Level of significance: 1%

Reject H_0 if $p\text{-value} \leq 0.01$

Step 7: Calculate the p -value directly from GC

Using GC, assume H_0 is true,

$p\text{-value} = 0.0569$ (3 s.f.) (refer to Table 2).

NORMAL FLOAT AUTO REAL RADIAN MP	NORMAL FLOAT AUTO REAL RADIAN MP
Z-Test Inpt:Data Stats μ_0 :1500 σ :.4 $\bar{x}$:1499.8 n :10 $\mu \neq \mu_0$ < μ_0 > μ_0 Color: BLUE Calculate Draw	Z-Test $\mu < 1500$ $z = -1.58113883$ $p = .0569231524$ $\bar{x} = 1499.8$ $n = 10$

Step 8: Conclude in context

Since $p\text{-value} = 0.0569 > 0.01$, we **do not reject** H_0 . Therefore, there is **insufficient evidence to conclude** that the machine underfills the bags at the 1% level of significance.

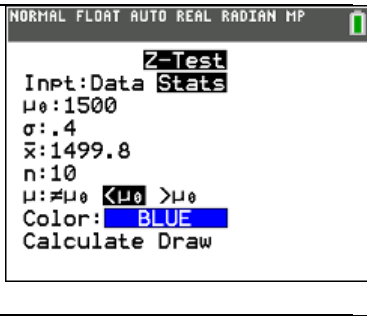
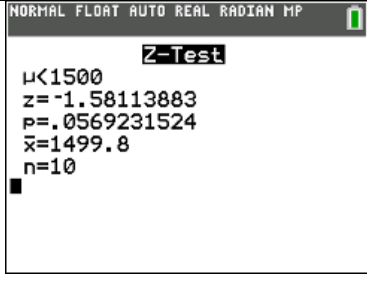
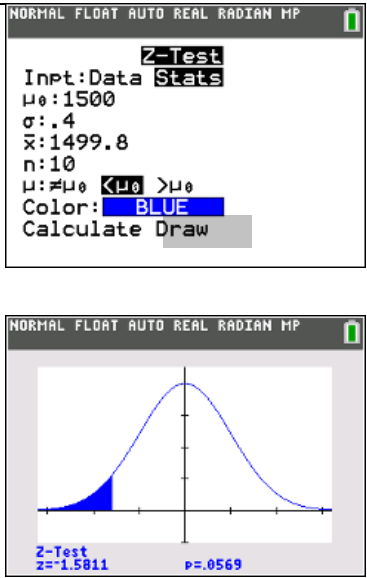
① Press [STAT] select TESTS, select 1: Z-Test, press [ENTER].	
② Key in the information as shown on the right. Note that we choose Stats for Inpt (input) because we are given the value of the sample mean, $\bar{x}$, rather than the raw data. Since we are testing $H_1: \mu < 1500$, we select $< \mu_0$ and press [ENTER]. Select 'Calculate' and press [ENTER].	
③ The screenshot shows the value of the test statistic, z, and the p-value, p.	
④ Note that if 'Draw' is selected, the region corresponding to the p-value is shaded in the normal distribution curve.	

Table 2

- (c) To conclude that the machine **underfills the bags**, we need to reject H_0 .

To reject H_0 , $p\text{-value} \leq \frac{\alpha}{100}$.

From part (a), $p\text{-value} = 0.0569$

i.e., $\frac{\alpha}{100} \geq 0.0569 \Rightarrow \alpha\% \geq 5.69\%$

Thus, the smallest level of significance = 5.7%. (1 d.p.)

Example 3

A machine is programmed to slice bread into slices with thickness that follows a normal distribution with mean 2.5 cm and standard deviation 0.2 cm. On a certain day, a sample of 80 slices of bread was found to have an average thickness of 2.45 cm. Test, at the 5% level of significance, whether the machine is functioning as programmed.

Solution

Step 1: Define the random variable and symbols

Let X be the thickness (in cm) of a slice of bread.

Let μ and σ^2 be the population mean and variance of X .

Step 2: State the data given

$\bar{x} = 2.45$ cm, $\mu = 2.5$ cm, $\sigma^2 = 0.2^2$, $n = 80$

Step 3: State the null and alternative hypotheses

To test $H_0: \mu = 2.5$ against $H_1: \mu \neq 2.5$

Note: this is a two-tailed test

Step 4: State the distribution of sample mean $\bar{X}$

Under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Step 5: Define the test statistic

Test statistic, $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$

Step 6: State the level of significance

Level of significance: 5%

Reject H_0 if $p\text{-value} \leq 0.05$

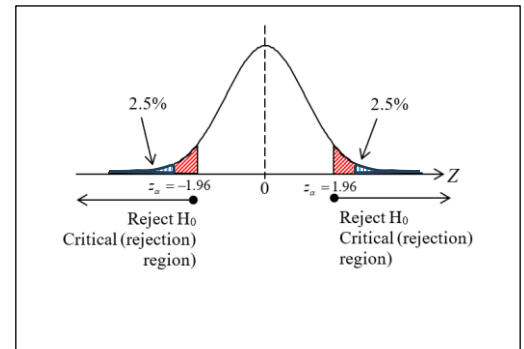
Step 7: Calculate the p -value

Using GC, assume H_0 is true,

$p\text{-value} = 0.0253$ (3 s.f.).

Step 8: Conclude in context

Since $p\text{-value} = 0.0253 < 0.05$, we reject H_0 at the 5% level of significance. Therefore, there is sufficient evidence to conclude that the machine is not functioning as programmed.



```
NORMAL FLOAT AUTO REAL RADIAN MP
Z-Test
Inpt:Data Stats
mu:2.5
sigma:0.2
x:2.45
n:80
mu:mu0 <mu0 >mu0
Color:BLUE
Calculate Draw
```

```
NORMAL FLOAT AUTO REAL RADIAN MP
Z-Test
mu=2.5
z=-2.236067977
p=.0253472347
x=2.45
n=80
```

Example 4

A machine is programmed to slice bread into slices with thickness that follows a normal distribution with mean 2.5 cm and standard deviation 0.2 cm. On a certain day, a sample of 80 slices of bread was found to have an average thickness of $\bar{x}$ cm. Find the critical region (i.e. set of values of $\bar{x}$) so that it can be concluded at the 10% level of significance that the machine is not functioning as programmed.

Solution

Step 1: Define the random variable and symbols

Let X be the thickness (in cm) of a slice of bread.

Let μ and σ^2 be the population mean and variance of X .

Step 2: State the data given

$\bar{x}$ is unknown, $\mu = 2.5$ cm, $\sigma^2 = 0.2^2$, $n = 80$

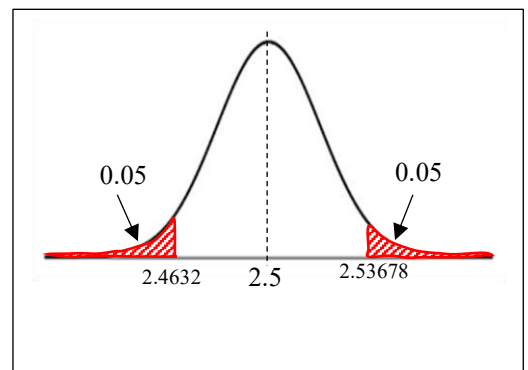
Step 3: State the null and alternative hypotheses

To test $H_0: \mu = 2.5$ against $H_1: \mu \neq 2.5$

Note: this is a two-tailed test

Step 4: State the distribution of sample mean $\bar{X}$

Under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.



Step 5: Find $\bar{x}$ given that H_0 is rejected at 10% level of significance.

Using GC, since there is sufficient evidence, at the 10% level of significance, to conclude that the machine is not functioning as programmed, therefore p -value ≤ 0.1 .

$$0 < \bar{x} \leq 2.46321 \quad \text{or} \quad \bar{x} \geq 2.53678$$

$$\therefore \text{The critical region is } \{\bar{x} \in \mathbb{R} : 0 < \bar{x} \leq 2.46 \quad \text{or} \quad \bar{x} \geq 2.54\}$$

⚡ Note:

Rejecting the null hypothesis, H_0 , means that we have enough evidence to support the alternative hypothesis, H_1 . However, not rejecting H_0 does not mean that H_0 is true - it only means that there is insufficient evidence to reject H_0 .

(B) #: Large sample from any population (not necessarily Normal)

If the population distribution is not Normal but the sample size n is sufficiently large, the Central Limit Theorem tells us that the sampling distribution of the sample mean $\bar{X}$ is approximately Normal, regardless of the original population distribution.

Under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately since n is sufficiently large.

Case 1: If the population variance σ^2 is **known**:

Test statistic, $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$ approximately

Case 2: If the population variance σ^2 is **unknown**,

The test statistic, $Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim N(0, 1)$ approximately,

where S^2 is the unbiased estimator of σ^2 (see Chapter S4).

Note:

1. The rule of thumb for ‘sufficiently large’ sample is $n \geq 30$.
2. The value s^2 can be calculated as follows (see Chapter S4):

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2,$$

$$s^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right),$$

$$s^2 = \frac{n}{n-1} \times \text{sample variance}.$$

3. **DO NOT** write $\bar{X} \sim N\left(\mu, \frac{S^2}{n}\right)$ as the parameters of a Normal distribution (i.e. the mean and variance) must be constants. As S^2 is computed from a random sample, its value will change from sample to sample – hence it is a random variable, not a constant.

Example 5

A company, FitTime, produces fitness trackers that measure heart rate using a new optical sensor. According to design specifications, the tracker’s readings should have a mean resting heart rate of 72 beats per minute (bpm) when tested on a standard calibration device.

To verify accuracy, engineers randomly selected 13 trackers and recorded their readings under controlled resting conditions:

70.8, 72.1, 73.0, 71.5, 72.4, 73.2, 71.8, 72.6, 72.9, 71.9, 72.3, 73.1, 72.5

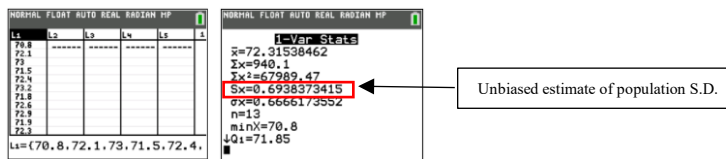
It is given that the population variance is ~~9~~ 0.5

- (a) Calculate the unbiased estimates of the population mean and population variance of the resting heart rate readings. [2]
- (b) Stating any necessary assumptions, test at the 10% level of significance whether the population mean resting heart rate differs from the expected value of 72 bpm. [6]
- (c) State the meaning of the p -value in this context. [1]
- (d) State, to the nearest percent, the smallest level of significance at which the null hypothesis would be rejected using the given set of data. [1]

Solution

- (a) Let X be the resting heart rate (bpm).

Since raw data is given in this question, we key into GC and use 1-Var Stats to find the sample mean and unbiased estimated of population variance.



Unbiased estimate of population mean = $\bar{x} = 72.31538$

Unbiased estimate of population variance = $s^2 = 0.693837^2$

- (b) Let μ and σ^2 be the population mean and variance of X .

$\bar{x} = 72.31538$, $\mu = 72$, $\sigma = \sqrt{0.5}$, $n = 13$

To test $H_0 : \mu = 72$ against $H_1 : \mu \neq 72$

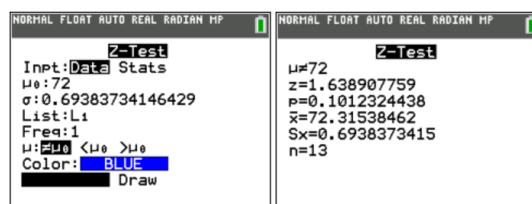
Assumption: Note that the **population distribution is unknown** and the **sample size is small** (Central Limit Theorem cannot be applied to approximate the distribution of $\bar{X}$), we need to assume that the population resting heart rate, X , is normally distributed. **i.e. Assume** $X \sim N(\mu, \sigma^2)$

Under H_0 , $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

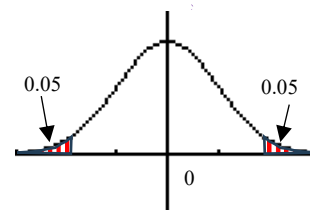
Test statistic, $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$

Level of significance: 10%

Reject H_0 if $p\text{-value} \leq 0.1$



Using GC, $p\text{-value} = 0.10123$ 0.1078



Since $p\text{-value} = 0.10123 > 0.1$, we do not reject H_0 at the 10% level of significance. Therefore, there is insufficient evidence to conclude that the population mean resting heart rate differs from the expected value of 72 bpm.

(c) The $p\text{-value}$ of 0.10123 is the probability of obtaining a sample mean resting heart rate of 72.4 bpm or higher, or 71.6 bpm or lower, if the population mean resting heart rate is in fact 72 bpm.

(d) H_0 is rejected if $p\text{-value} \leq \frac{\alpha}{100}$, i.e. $\frac{\alpha}{100} \geq 0.10123$ 0.1078

Hence smallest level of significance to the nearest percent is ~~11%~~ 10.8%

Example 6

Clara, who has diabetes, monitors her blood glucose levels, which vary throughout the week. A random sample of 75 readings (x) is taken over a week, are statistics summarised by

$$\sum x = 511.5 \text{ and } \sum x^2 = 4027.89.$$

Find the set of values of μ_0 for which $\mu > \mu_0$ is accepted at the 10% level of significance.

Solution

Let X be the blood glucose level of Clara.

Let μ and σ^2 be the population mean and variance of X .

Note that population distribution is not Normal and population variance is unknown.

$$\bar{x} = \frac{\sum x}{n} = \frac{511.5}{75} = 6.82, \mu = \mu_0, n = 75$$

$$s^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{75-1} \left(4027.89 - \frac{(511.5)^2}{75} \right) = 7.29$$

To test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$

Under H_0 , since n is large, then by Central Limit Theorem, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately.

$$\text{Test statistic, } Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim N(0, 1) \text{ approximately}$$

Level of significance : 10%

Method 1 (By GC)

Reject H_0 if $p\text{-value} \leq 0.1$, i.e. $P(\bar{X} > 6.82) \leq 0.1$

Let $Y_1 = P(\bar{X} > 6.82)$ and $Y_2 = 0.1$



From GC $\mu_0 \leq 6.42045$

$$\{\mu_0 \in \mathbb{R} : 0 < \mu_0 \leq 6.42\}$$

Method 2 (By standardisation)

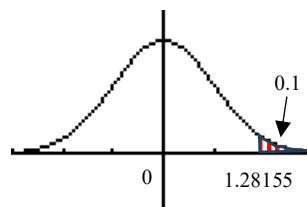
Reject H_0 if $p\text{-value} \leq 0.1$, i.e. $z \geq 1.28155$

$$z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \geq 1.28155$$

$$\frac{6.8200 - \mu_0}{\sqrt{\frac{7.2900}{75}}} \geq 1.28155$$

$$\mu_0 \leq 6.42045$$

$$\{\mu_0 \in \mathbb{R} : 0 < \mu_0 \leq 6.42\}$$



Note (Rounding in Hypothesis Testing):

Always round your final answer in a way that does not change the conclusion of the hypothesis testing.

For example, if your calculation gives

$$\mu_0 \leq 6.42545,$$

you should state the answer as

$$0 < \mu_0 \leq 6.42,$$

and not $0 < \mu_0 \leq 6.43$.

This is because rounding up may cause the $p\text{-value}$ to drop below 0.1, which would contradict the decision to accept H_0 .

Example 7

A machine is designed to fill sacks with rice such that the mass of each sack has a mean of 350 g and a standard deviation of 20 g. To check if the machine's settings have changed, a random sample of 60 filled sacks is taken, and the mean mass is found to be 356 g.

- Test, at the 5% level of significance, whether the machine's settings have changed.
- Using the same sample, explain whether there is sufficient evidence to conclude that:
 - the machine's settings have changed at the 10% level of significance;
 - the mean mass of a sack of rice has increased at the 1% level of significance.

Solution

- (a) Let X be the mass (in grams) of a sack of rice.

Let μ and σ^2 be the population mean and variance of X .

$$\bar{x} = 356, \mu = 350, \sigma = 20, n = 60$$

To test $H_0 : \mu = 350$ against $H_1 : \mu \neq 350$

Under H_0 , since n is large, then by Central Limit Theorem, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

approximately.

Test statistic, $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$ approximately.

Level of significance: 5%

Reject H_0 if $p\text{-value} \leq 0.05$

Assuming H_0 is true, using GC, $p\text{-value} = 0.0201$ (3 s.f.).

Since $p\text{-value} = 0.0201 < 0.05$, we reject H_0 at the 5% level of significance. There is sufficient evidence to conclude that the settings of the machine have changed.

- (i) Since there is sufficient evidence to say that the settings of the machine have changed at the 5% level of significance, the $p\text{-value}$ for this 2-tailed test is less than 0.05, which in turn is less than 0.1, i.e. $p\text{-value} < 0.05 < 0.1$. Hence there is sufficient evidence to conclude that the settings of the machine have changed at the 10% level of significance.
- (ii) To test $H_0 : \mu = 350$ against $H_1 : \mu > 350$ at the 1% level of significance.

For this right-tailed test, the new $p\text{-value} = \frac{0.0201}{2}$.

Since the new $p\text{-value} > 0.01$, we do not reject H_0 at the 1% level of significance. Hence there is insufficient evidence to conclude that the mean mass of a sack of rice has increased.

Note

1. If a test result is significant at the 5% level of significance, it will also be significant at any higher level of significance (for example, 6% or 10%).
2. When using the same sample, the value of the test statistic remains the same for both one-tailed and two-tailed tests, since it is calculated using the same formula for $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$.

However, the $p\text{-value}$ for a two-tailed test is **twice** that of the corresponding one-tailed test.

Summary of Hypothesis Tests for a Population Mean μ (1-Sample z-Test)

Choice of Test Statistic		
Population Distribution	σ^2 Variance Known	σ^2 Variance Unknown
Normal $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$	$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$	$Z = \frac{\bar{X} - \mu_0}{\frac{\hat{\sigma}}{\sqrt{n}}} \sim N(0,1)$ approx
Not Normal / Unknown $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ approx. (large n by CLT)	$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ approx.	$Z = \frac{\bar{X} - \mu_0}{\frac{\hat{\sigma}}{\sqrt{n}}} \sim N(0,1)$ approx.

Notes:

- For **small samples** from a non-normal population: a 1-sample **z-test is not valid**.
- For **small samples** from a normal population with **unknown variance**, we use a **t-test** (not covered in H2 Math syllabus).

Steps in Conducting a Hypothesis Test:

- Define the random variable, X , and identify population mean and variance, μ and σ^2 of the random variable.
- Set up the hypotheses:
 - Null Hypothesis H_0
 - Alternative Hypothesis H_1 (one-tailed or two-tailed)
- State sample information:
 - Sample size n
 - Sample mean $\bar{x}$
 - Known variance σ^2 or unbiased estimate of population variance s^2
- Under H_0 , specify the distribution of the test statistic, $\bar{X}$:
 - If X is normally distributed, use the Z-test directly
 - If distribution of X is unknown and n is large, use **Central Limit Theorem** justify the approximate Z test
- Choose one of the two testing methods
 - p-value approach**
 - Specify the level of significance
 - Compute the test statistic value and the corresponding p -value
 - Compare: reject H_0 if $p\text{-value} \leq \frac{\alpha}{100}$
 - Critical (rejection) region approach**
 - Specify the level of significance
 - Find the critical value(s) and define the critical (rejection) region
 - Reject H_0 if the test statistic value falls inside the critical (rejection) region
- Make a Conclusion

- State whether you reject or do not H_0
- Give the conclusion in the **context** of the question.

Statistical Tutorial 5 Hypothesis Testing

1. A Level/N17/02/Q7 (modified)

The production manager of a food manufacturing company wishes to take a random sample of a certain type of biscuit bar from the thousands produced one day at his factory, for quality control purposes. He wishes to check that the mean mass of the bars is 32 grams, as stated on the packets.

- (i) Explain why a random sample is needed for the production manager to conduct this test. [1]

The masses, x grams, of a random sample of 40 biscuit bars are summarised as follows.

$$n = 40 \quad \sum(x - 32) = -7.7 \quad \sum(x - 32)^2 = 11.05$$

- (ii) Calculate the unbiased estimates of the population mean and variance of the mass of the biscuit bars. [2]
- (iii) Test, at 1% level of significance, the claim that the mean mass of biscuit bars is 32 grams. You should state your hypotheses and define the symbols you use. [5]
- (iv) Explain, in the context of the question, the meaning of “at 1% level of significance” for the test in (iii). [1]
- (v) Explain why there is no need for the production manager to know anything about the distribution of the masses of the biscuit bars. [2]

2. Y1JC/Prelim 9758/2019/02/Q10(i)to(iii) (modified)

A previous study revealed that the average time taken to assemble a certain type of electrical component is at least 15 minutes. The manager wants to investigate if the result of the study is valid. A random sample of 40 components is taken and the times taken to assemble the components are summarised in the following table:

Time to assemble a component (min)	9	10	12	13	15	16	17	18
Number of components	1	6	3	8	5	7	8	2

- (i) Explain what it means for a sample to be random in this context and why a random sample is needed. [2]
- (ii) Find unbiased estimates of the population mean and variance. [2]
- (iii) What do you understand by the term “unbiased estimator”? [1]
- (iv) Test at the 5% level of significance whether the results of the study is valid. You should state your hypotheses and define any symbols you use. [5]
- (v) Explain why the manager is able to conduct the test without knowing anything about the distribution of the times taken to assemble the electrical components. [1]

3. **DHS Prelim 9758/2022/02/Q9**

- (a) A confectionery is selling mooncakes during the Mid-Autumn Festival. The owner claims that the mean mass of the mooncakes is at least 150 g. The distribution of the mass of a mooncake has standard deviation 6.73 g. A customer bought a random selection of 9 mooncakes from the confectionery with masses, in g, as follows

145	148	153	156	141	151	143	157	138
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Test at the 10% significance level whether the owner's claim is valid. State an assumption about the population distribution of the mass of the mooncakes. [5]

- (b) In a particular country, the mean working hours of teachers in schools is 60 hours per week. After implementation of the Home-Based Learning in all schools, the Ministry claims that the mean working hours of teachers remained unchanged. A random sample of 50 teachers was taken and the number of working hours per week was recorded. The mean working hours and standard deviation for the sample were found to be 62 hours and k hours respectively. A hypothesis test was conducted and it was found that there is sufficient evidence to reject the Ministry's claim at 5% significance level. Find the set of values of k . [4]

4. **SAJC Prelim 9758/2024/02/Q11 (modified)**

- (a) Ms Lam, a teacher from Shelton College, claims that students spend an average of 4 hours studying in the college per week. A survey was conducted and the amount of time, in hours, spent by 50 randomly chosen students studying in the college were recorded. It was found that the amount of time that the students spent studying in the college per week, in hours, had a mean of 3.8 hours and a standard deviation of 0.46 hours.
- (i) Test, at the 5% level of significance, whether the average amount of time spent by students studying in the college per week is 4 hours. [5]
- (ii) Explain the meaning of the p -value obtained in (i) in the context of the question. [1]
- (iii) Explain if the test in part (i) would still be valid if the amount of time spent by students studying in the college was collected from a random sample of 15 students instead. [1]
- (b) A publicity campaign on the night study program is held. Ms Lam claims that due to the publicity campaign, the mean amount of time that students spend studying in the college is now μ_0 hours. Another survey on 8 students was conducted after the publicity campaign, and the amount of time, in hours, spent studying in the college for each student were recorded as follows:

4.3 4.5 3.8 4.4 4.9 3.5 4.6 3.9

It is now known that the amount of time that students spend studying in Shelton College after the publicity campaign follows a normal distribution with a standard deviation of 0.6 hours.

- (i) Calculate the sample mean amount of time spent studying in Shelton College of the 8 students. [1]
- (ii) Find the largest possible value of μ_0 , such that at 5% level of significance, there is insufficient evidence that Ms Lam has overstated the mean amount of time that the students spent studying in Shelton College. [4]
- (iii) Explain, in the context of the question, the meaning of “at 5% level of significance”. [1]

5. Specimen Paper/2017/02/Q10

The average time required for the manufacture of a certain type of electronic control panel is 17 hours. An alternative manufacturing process is trailed, and the time taken, t hours, for the manufacture of each of 50 randomly chosen control panels using the alternative process is recorded. The results are summarized as follows.

$$n = 50 \quad \sum t = 835.7 \quad \sum t^2 = 14067.17$$

The Production Manager wishes to test whether the average time taken for the manufacture of a control panel is different using the alternative process, by carrying out a hypothesis test.

- (i) Explain whether the Production Manager should use a 1-tail test or a 2-tail test. [1]
- (ii) Explain why the Production Manager is able to carry out a hypothesis test without knowing anything about the distribution of the times taken to manufacture the control panels. [2]
- (iii) Find the unbiased estimates of the population mean and variance and carry out the test at the 10% level of significance for the Production Manager. [6]
- (iv) Suggest a reason why the Production Manager might be prepared to use an alternative process that takes a longer average time than the original process. [1]

The Finance Manager wishes to test whether the average time taken for the manufacture of a control panel is shorter using the alternative process. The Finance Manager finds that the average time taken for the manufacture of each of 40 randomly chosen control panels, using the alternative process, is 16.7 hours. He carries out a hypothesis test at the 10% level of significance.

- (v) Explain, with justification, how the population variance of the times will affect the conclusion made by the Finance Manager. [3]

6. **EJC Prelim 9758/2024/02/Q11 (modified)**

The heel stack of a running shoe refers to the thickness of the sole of the shoe at the point where the heel of the foot sits in the shoe.

A running shoe manufacturer makes a model called “SuperFly” and claims that the average heel stack is 42.7 millimetres. After reading some online reviews, the production manager wishes to test if the heel stack of the Superfly has been overstated.

- (a) Explain whether the manager should carry out a 1-tail or 2-tail test. State appropriate hypotheses for the test, defining any symbols you use. [3]

The production manager goes to the factory to measure a random sample of 16 shoes. The heel stacks, x millimetres, are summarised as follows.

$$n=16 \quad \sum(x-42.7)=-6.4 \quad \sum(x-42.7)^2=13.2$$

- (b) State ~~two~~^{ar} assumptions the production manager need to make to carry out the hypothesis test in part (a). [2]
- (c) Calculate unbiased estimates for the population mean and population variance of the heel stacks, and carry out the hypothesis test in part (a) at 2.5% level of significance. [5]

The manufacturer upgrades the machine used to produce the sole of the Superfly. It is later established that the heel stack of the Superfly has a variance of 0.987 millimetres². The production manager tests another random sample of 20 shoes to see whether the mean heel stack has changed, at 2.5% level of significance.

- (d) With the same assumption(s) as in part (b), find the range of values of the sample mean, in millimetres to 1 decimal place, such that he can conclude that the mean heel stack has changed. (Answers obtained by trial and improvement from a calculator will obtain no marks.) [3]

7. **ACJC Prelim 9758/2023/02/Q11**

Energy bars are a convenient source of nutrition for athletes. A typical energy bar weighs between 30-50 g and is likely to supply about 200-300 calories. An American company sells energy bars and a nutritionist working in the company claims that the mean number of calories in a product, Perfect Protein bars, is 350 calories.

A random sample of 30 bars is collected and the number of calories, x cal, in the 30 bars is summarised as follows.

$$\sum x = 10446, \sum x^2 = 3638000.$$

- (i) Test at the 3% level of significance whether the mean number of calories in Perfect Protein bars is less than 350 calories, defining any symbols that you use. [7]
- (ii) Explain why the test would be inadmissible if the nutritionist had taken a random sample of 15 energy bars and hence state an assumption needed for him to carry out the test. [1]

- (iii) If the test was for whether the mean number of calories in Perfect Protein bars is 350 calories, find the smallest level of significance for which the nutritionist's claim will be rejected. [2]

The company launched a new product called Dynamic Protein bars which are advertised to be higher in calories than Perfect Protein bars. A random sample of 45 Dynamic Protein bars is collected to test this and the mean number of calories of the sample is found to be 352.2.

- (iv) The result of this investigation shows that there is no reason to believe that the mean number of calories in the new product is more than 350 calories at the 4% level of significance. Find the range of possible values of the variance used in calculating the test statistic. [3]

8. **EJC Prelim 9758/2021/02/Q9**

A company produces light bulbs with a brightness rating of 470 lumens. It is known that the brightness of these light bulbs is normally distributed with a standard deviation of 10 lumens. A production manager wishes to test whether the mean brightness is less than 470 lumens. He collects a random sample of 40 light bulbs and finds that their mean brightness is 466.8 lumens.

- (i) Explain what is meant by the term “random sample” in the context of this question and why is it necessary. [2]
- (ii) Carry out a test at the 5% level of significance for the production manager, clearly stating the p -value of this test. Explain what p -value means in the context of this question. [5]
- (iii) Suppose that the production manager had tested if the mean brightness differed from 470 lumens at the 5% level of significance. Without performing another hypothesis test, determine whether your conclusion in **part (ii)** would be affected. [2]

The company decides to produce a new line of ‘lowlight’ bulbs with a brightness rating of 350 lumens. A random sample of 50 ‘lowlight’ bulbs is taken and the brightness level, y lumens, are summarised as follows.

$$n = 50 \qquad \sum (y - 350) = -125 \qquad \sum (y - 347.5)^2 = 4704$$

- (iv) Calculate unbiased estimates of the population mean and variance for the brightness of the ‘lowlight’ bulbs. [2]
- (v) Given instead that the standard deviation of the brightness of the ‘lowlight’ bulbs is known to be σ , the company found insufficient evidence from this sample to conclude that the mean brightness was less than 350 lumens at the 1% level of significance. Determine the range of values of σ . [2]

9. **RI Prelim 9758/2024/02/Q10**

- (a) A company has a machine designed to fill bags with, on average, μ_0 kg of salt. The mass of salt in a randomly chosen bag has a normal distribution with population standard deviation denoted by σ kg. The production manager wishes to investigate if the machine is adjusted correctly. He takes a random sample of n bags and carries out a hypothesis test at the 1% level of significance.
- (i) State null and alternative hypotheses for the manager's test, defining any parameters you use. [2]
- (ii) Find, in terms of μ_0 , σ and n , the critical region(s) for this test. [4]
- (b) The company has a different machine which fills bags with, on average, 25 kg of low sodium salt. One of the company's production supervisors has reported that some of the workers suspect the machine is no longer set correctly, and the average mass of low sodium salt in the bags may in fact be more than 25 kg. The production supervisor decides to carry out a hypothesis test at the 0.5% level of significance with a random sample of 80 bags of low sodium salt. Summary data for the mass, y kg, of low sodium salt in these bags is as follows.

$$n = 80 \quad \sum (y - 25) = 27.2 \quad \sum (y - 25)^2 = 85.1$$

- (i) Carry out the test and state the conclusion of the test in the context of the question. [5]
- (ii) State, giving a reason, whether any assumptions about the population are needed in order for the test to be valid. [1]

10. **ACJC Prelim 9758/2022/02/Q10**

A nasi lemak stall holder uses fresh chicken wings as an ingredient for fried chicken wings. Based on his past years records, his mean daily profit was \$535. With the recent lack of fresh chicken supply, the stall holder substituted fresh chicken wings with frozen chicken wings as the ingredient. His wife was hesitant to the change and claimed that the mean daily profit will decrease. To test his wife's claim, the stall holder takes a random sample of 45 days and recorded the daily profits, $\$x$.

- (i) State appropriate hypotheses to test the wife's claim and define any symbols that you use. [2]
- (ii) State, with a reason, whether it is necessary to assume that his past years records of daily profits are normally distributed for the test to be valid. [1]
- (iii) Based on the past years records, it is assumed that the population variance of the daily profit is 2591. If the test shows that there is sufficient evidence that the wife's claim is accepted at 5% level of significance, determine the set of possible values of $\bar{x}$, the mean daily profit in the 45 days. [2]

- (iv) The stall holder found that $\bar{x} = \$520$ and suspects that the population variance of 2591 may be incorrect. Hence he decided to use the sample variance value of 2008 to test his wife's claim. State the conclusion of the test, showing your workings clearly. [3]
- (v) State the largest significance level that the stall holder should use so that the conclusion in (iv) will be different. Leave your answer in 2 decimal places. [1]

The stall holder now suspects that the mean daily profit does not differ from \$535, even if frozen chicken wings are used instead of fresh chicken wings. To test his claim, the stall holder decides to increase the number of randomly selected days, n , to record his daily profit.

- (vi) It is given that $\bar{x} = 526$ and the population variance is assumed to be 2591. Determine the greatest value of n , so that the conclusion of the test shows that there is no reason to reject the null hypothesis at 8% level of significance. [4]

Answers

1. (ii) $\frac{12723}{400}$; 0.245
2. (ii) 14.2; 7.09
3. (b) $\{k \in \mathbb{R} : 0 < k \leq 7.14 \text{ or } k \geq 7.15\}$
4. (bi) 4.2375 (ii) 4.58 h
5. (iii) 16.714; 2.03
6. (c) 42.3; 0.709 (d) $\bar{x} \leq 42.2$ or $\bar{x} \geq 43.2$
7. (iii) 4.53% (iv) $\sigma^2 > 71.1$
8. (iv) 347.5; 96 (v) $\sigma \geq 7.60$
9. $\left(0, \mu_0 - \frac{2.58\sigma}{\sqrt{n}}\right] \cup \left[\mu_0 + \frac{2.58\sigma}{\sqrt{n}}, \infty\right)$
10. (iii) $\{\bar{x} : 0 < \bar{x} < \frac{522}{523}\}$ (v) 1.31% (vi) 98

522
round off in context

Skill Set

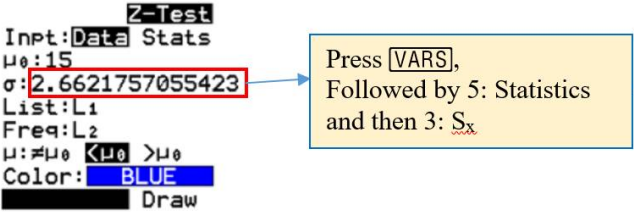
Chapter S5 Hypothesis Testing

No.	Skills	Examples of questions involving the skills
1	State the definitions of (a) null and alternative hypotheses, (distinguish between 1-tailed and 2-tailed tests) (b) test statistic, (c) critical (rejection) region and critical value, (d) level of significance and p-value.	Refer to Statistical Chapter 5 Notes for detailed definitions. [Lecture Notes Example 1] In the following situations, state suitable null and alternative hypotheses. (a) The mean factory assembly time for a particular electronic component is 84 seconds. The manager wants to test whether the new process results in a different assembly time. (b) The reported average age of hospital patients is 53 years. The director wants to test whether this figure is an underestimate. Solution: (a) Let μ be the population mean factory assembly time (in seconds). To test $H_0: \mu = 84$ against $H_1: \mu \neq 84$ (b) Let μ be the population average age (in years) of the hospital patients. To test $H_0: \mu = 53$ against $H_1: \mu > 53$ [Tutorial Q1(iv)] Explain, in the context of the question, the meaning of “at 1% level of significance” for the test in (iii). Solution: “1% level of significance” means that the probability of wrongly concluding that the population mean mass of biscuit bars is not 32g when it is indeed 32g is 0.01. [Tutorial Q4(ii)] Explain the meaning of the p-value obtained in (i) in the context of the question

No.	Skills	Examples of questions involving the skills
		Solution: p -value is the probability of observing the sample mean time a randomly chosen sample of 50 students spent studying in the college per week to be equal or less than 3.8 hours or to be equal or more than 4.2 hours, when the population mean time spend by students studying in the college per week is indeed 4 hours is 0.00234.
2	Formulate hypotheses and test for a population mean based on - a sample taken from a normal population with known or unknown variance, and - a large sample taken from any population (including normal distribution) with the use of the Central Limit Theorem.	- Define the random variable in the population, the population mean and population standard deviation, - Write out the null and alternative hypotheses. Do not write $H_0 = 170$. It is not the null hypothesis that is equal to 170. Rather, 'the population mean is equal to 170' is the null hypothesis. - List out the sample size, sample mean and the population variance (if given) Write out the distribution for the Sample Mean (Check if use of Central Limit Theorem required) - Write out the Test Statistic - Use either p-value from the GC or - compare the z-value (test statistic value) with the critical value(s) - Make your conclusion in context to the question
3	Questions involving claim that is described in the form of an inequality	A previous study revealed that the average time taken to assemble a certain type of electrical component is at least 15 minutes. In this case, the null hypothesis is $\mu \geq 15$ however we will write it as $\mu = 15$ $H_0 : \mu = 15$ [$\mu \geq 15$] And the alternative hypothesis will be $H_1 : \mu < 15$

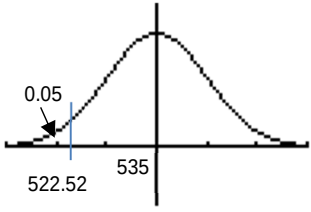
No.	Skills	Examples of questions involving the skills
4a)	Determine when to use the Central Limit Theorem	[Tutorial Q4a] Ms Lam, a teacher from Shelton College, claims that students spend an average of 4 hours studying in the college per week. A survey was conducted and the amount of time, in hours, spent by 50 randomly chosen students studying in the college were recorded. It was found that the amount of time that the students spent studying in the college per week, in hours, had a mean of 3.8 hours and a standard deviation of 0.46 hours. Solution: Under H_0, since $n = 50$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{50}\right)$ approximately
		[Tutorial Q4b] It is now known that the amount of time that students spend studying in Shelton College after the publicity campaign follows a normal distribution with a standard deviation of 0.6 hours. Solution: Under H_0, $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{8}\right)$, i.e. $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ Unlike Question 4a, in Question 4b we were told that X follows a normal distribution, so $\bar{X}$ would follow a normal distribution as well. There is NO NEED to use Central Limit Theorem here.

No.	Skills	Examples of questions involving the skills
4b)	Determine the use of population variance (σ^2) or unbiased estimate of population variance (s^2)	[Tutorial Q4a]: Ms Lam, a teacher from Shelton College, claims that students spend an average of 4 hours studying in the college per week. A survey was conducted and the amount of time, in hours, spent by 50 randomly chosen students studying in the college were recorded. It was found that the amount of time that the students spent studying in the college per week, in hours, had a mean of 3.8 hours and a standard deviation of 0.46 hours. Solution: Population variance is unknown. Unbiased estimate of the population variance $s^2 = \frac{50}{49}(0.46)^2 = 0.21592 \approx 0.216 \text{ (3 sig. fig.)}$ Under H_0, since sample size = 50 is large, by Central Limit Theorem, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{50}\right)$ approximately, $Z = \frac{\bar{X} - \mu}{\sqrt{\frac{s^2}{40}}} \sim N(0, 1) \text{ approximately}$
		[Tutorial Q4b] It is now known that the amount of time that students spend studying in Shelton College after the publicity campaign follows a normal distribution with a standard deviation of 0.6 hours. Unlike Question 4a, in Question 4b, we were given information on the population variance. Hence we can use the population variance. To test $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ at 5% level of significance Under H_0, $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{8}\right)$,

No.	Skills	Examples of questions involving the skills
		i.e. $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ In short, our first choice for variance is population variance, σ^2 but if that is not available, then we use the unbiased estimate for population variance, s^2.
5	Questions where the sample data is given in the form of a table - Use GC - Z-TEST – Data - Key in Data in L₁ and L₂ using EDIT - Compute 1 Var Stats Perform Z-Test	[Tutorial Qn 2] A previous study revealed that the average time taken to assemble a certain type of electrical component is at least 15 minutes. The manager wants to investigate if the result of the study is valid. A random sample of 40 components is taken and the times taken to assemble the components are summarised in the following table ... Solution: 
6	Given the conclusion, find range of values for unknown by comparing test statistic value with critical value(s) note on how to round off inequalities for questions with a context	[Tutorial Q3b] sample standard deviation is unknown In a particular country, the mean working hours of teachers in schools is 60 hours per week. After implementation of the Home-Based Learning in all schools, the Ministry claims that the mean working hours of teachers remained unchanged. A random sample of 50 teachers was taken and the number of working hours per week was recorded. The mean working hours and standard deviation for the sample were found to be 62 hours and k hours respectively. A hypothesis test was conducted and it was found that there is sufficient evidence to reject the Ministry's claim at 5% significance level. Find the set of values of k. Solution: $s^2 = \frac{n}{n-1} (\text{sample variance}) = \frac{50k^2}{49} \text{ hours}^2$ To test $H_0: \mu = 60$ again $H_1: \mu \neq 60$

No.	Skills	Examples of questions involving the skills
		Under H_0, $\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately by Central Limit Theorem since sample size of 50 is large. Test statistic $Z = \frac{\bar{X} - \mu}{\sqrt{\frac{S^2}{n}}} \sim N(0,1)$ approximately In order to reject H_0, $p\text{-value} \leq 0.05$ $0 < k \leq 7.14299$ or $k \geq 7.14299$ Set of values of k is $\{k \in \mathbb{R} : 0 < k \leq 7.14 \text{ or } k \geq 7.15\}$ [Tutorial Q4 (bii)] population mean is unknown A publicity campaign on the night study program is held. Ms Lam claims that due to the publicity campaign, the mean amount of time that students spend studying in the college is now μ_0 hours. Another survey on 8 students was conducted after the publicity campaign, and the amount of time, in hours, spent studying in the college for each student were recorded as follows: Find the largest possible value of μ_0, such that at 5% level of significance, there is insufficient evidence that Ms Lam has overstated the mean amount of time that the students spent studying in Shelton College. [4] Solution: To test $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ at 5% level of significance Under H_0, $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{8}\right)$, i.e. $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{8}}} \sim N(0,1)$ Since there is insufficient evidence that Ms Lam has overstated the average time, we do not reject H_0 at 5% (z-calc does not lie in critical region) By GC: $p\text{-value} > 0.05$ $\mu_0 < 4.5864$

No.	Skills	Examples of questions involving the skills
		Largest possible value of μ_0 is 4.58 h. [Tutorial Q7(iv)] population variance is unknown The company launched a new product called Dynamic Protein bars which are advertised to be higher in calories than Perfect Protein bars. A random sample of 45 Dynamic Protein bars is collected to test this and the mean number of calories of the sample is found to be 352.2. The result of this investigation shows that there is no reason to believe that the mean number of calories in the new product is more than 350 calories at the 4% level of significance. Find the range of possible values of the variance used in calculating the test statistic. [3] Solution: To test $H_0 : \mu = 350$ against $H_1 : \mu > 350$ At 4% level of significance, we do not reject H_0. For p-value > 0.04 By GC: $\sigma^2 > 71.0626$ $\therefore \sigma^2 > 71.1$ (3 sf.)
		[Tutorial Q10 (iii)] sample mean is unknown A nasi lemak stall holder uses fresh chicken wings as an ingredient for fried chicken wings. Based on his past years records, his mean daily profit was \$535. With the recent lack of fresh chicken supply, the stall holder substituted fresh chicken wings with frozen chicken wings as the ingredient. His wife was hesitant to the change and claimed that the mean daily profit will decrease. To test his wife's claim, the stall holder takes a random sample of 45 days and recorded the daily profits, \$x. Based on the past years records, it is assumed that the population variance of the daily profit is 2591. If the test shows that there is sufficient evidence that the wife's claim is accepted at 5% level of significance, determine the set of possible values of $\bar{X}$, the mean daily profit in the 45 days. [2] Solution: Since H_0 is rejected at 5% level of significance,

No.	Skills	Examples of questions involving the skills
		By GC, $\bar{x} = 522.52$ $\therefore \{ \bar{x} : 0 < \bar{x} < 523 \}$ 
7	Using p -value to find the range of values of level of significance	[Tutorial Q10 (v)] State the largest significance level that the stall holder should use so that the conclusion in (iv) will be different. Leave your answer in 2 decimal places. [1] Solution If H_0 is not rejected, $p\text{-value} > \frac{\alpha}{100}$ $0.013195 > \frac{\alpha}{100}$ $\alpha < 1.3195$ The largest level of significance is 1.31%. (to 2 d.p)
8	Stating assumptions	[Tutorial Q10(ii)] large sample size, no need to assume X follows a normal distribution. A nasi lemak stall holder uses fresh chicken wings as an ingredient for fried chicken wings. Based on his past years records, his mean daily profit was \$535. With the recent lack of fresh chicken supply, the stall holder substituted fresh chicken wings with frozen chicken wings as the ingredient. His wife was hesitant to the change and claimed that the mean daily profit will decrease. To test his wife's claim, the stall holder takes a random sample of 45 days and recorded the daily profits, \$$x$. Solution: It is not necessary to assume that the daily profits are normally distributed for the test to be valid since $n = 45$ is large, by Central Limit Theorem the mean daily profits are approximately normally distributed.

No.	Skills	Examples of questions involving the skills
		[Tutorial Q7] small sample size, so need to assume X follows normal distribution Explain why the test would be inadmissible if the nutritionist had taken a random sample of 15 energy bars and hence state an assumption needed for him to carry out the test. [1] Solution: The test would be inadmissible if the nutritionist had taken a random sample of 15 energy bars as the distribution of the population is unknown and Central Limit Theorem cannot be applied to approximate the distribution of the sample mean number of calories to be normal as the sample size is small. The nutritionist will need to assume that the number of calories in Perfect Protein bars follow a normal distribution.

Supplementary Exercise

1. 2017/MJC/BT2/I/Q8

A factory produces batteries. The lifetime of a battery is X hours, and the results from a randomly chosen sample of 100 batteries are summarised by

$$\sum x = 1150, \text{ and } \sum x^2 = 15680.$$

- (i) Find unbiased estimates of the population mean and variance. [2]
- (ii) The factory claims that the mean lifetime of the batteries is at least 12 hours. Test at 1% level of significance if the factory's claim is justified. [5]

2. (2017/DHS/BT2/II/Q1)(modified)

A cereal manufacturer produces thousands of packets of cereal each day. The mass of cereal in one packet is denoted by X grams. Based on past records, the mass of cereal in a randomly chosen packet has a mean of 470 g.

Following a slight adjustment to the packing machine, the masses of cereal in 30 randomly chosen packets are summarized by

$$\sum x = 13999, \sum x^2 = 6534258.$$

The manufacturer wishes to find out if the mean mass of cereal is different after making the adjustment, by carrying out a hypothesis test.

- (i) Explain whether the manufacturer should use a 1-tail test or 2-tail test. [1]
- (ii) Explain why the manufacturer is able to carry out a hypothesis test without knowing anything about the distribution of mass of cereal in the cereal packets. [2]
- (iii) Calculate the unbiased estimates of the population mean and variance and carry out the test at the 5% significance level for the cereal manufacturer. [6]

3. A Level/2019/02/Q7

The production manager of a food manufacturing company wishes to take a random sample of a certain type of biscuit bar from the thousands produced one day at his factory, for quality control purposes. He wishes to check that the mean mass of the bars is 32 grams, as stated on the packets.

- (i) State what it means for a sample to be random in this context. [1]

The masses, x grams, of a random sample of 40 biscuit bars are summarized as follows.

$$n = 40 \qquad \sum (x - 32) = -7.7 \qquad \sum (x - 32)^2 = 11.05$$

- (ii) Calculate unbiased estimates of the population mean and variance of the mass of biscuit bars. [2]

- (iii) Test, at the 1% level of significance, the claim that the mean mass of biscuit bars is 32 grams. You should state your hypotheses and define any symbols you use. [5]
- (iv) Explain why there is no need for the production manager to know anything about the population distribution of the masses of the biscuit bars. [2]

4. **NJC Prelim 9758/2024/02/Q9**

A factory manufactures cans and bottles of iced tea. Machine A is used to fill the cans with iced tea and Machine B is used to fill the bottles with iced tea. Machine A is set to fill each can with 300 millilitres (ml) of iced tea. A random sample of 60 filled cans of iced tea was taken and the volume, x ml, of iced tea in each can was measured. The following summarised data was obtained.

$$\sum(x-300) = -112.8, \quad \sum(x-300)^2 = 4532.87$$

- (a) Defining clearly any symbols you use, test at the 8% level of significance, whether the mean volume of iced tea per can is 300 ml. [6]
- (b) Explain in the context of the question, the meaning of ‘at the 8% level of significance’. [1]
- (c) The manager of the factory claims that the mean volume of iced tea that Machine B fills per bottle with is at least 500 ml. It is found that the volumes of iced tea in the bottles filled by Machine B follow a normal distribution with standard deviation 5 ml. A random sample of 35 filled bottles was taken and a test for the validity of the manager’s claim was carried out at the 4% level of significance. Find the critical region for this test, correct to 1 decimal place. [3]

5. **ASRJC Prelim 9758/2021/02/Q11**

An airline which caters various flights within the United States of America has a flight route that travels from New Orleans to Miami. The airline claims that the average flight time is 115 minutes. Brandon, a regular commuter of this flight route, thinks that his journey takes a shorter time on average.

- (i) Explain, whether Brandon should carry out a 1-tail test or a 2-tail test. State the hypotheses for the test, defining any symbols you use. [2]

Brandon records the flight time (in minutes) of 8 randomly selected flights as shown below.

113.0 112.4 110.0 113.8 111.3 115.2 114.2 115.5

- (ii) Assuming that the flight time from New Orleans to Miami follows a normal distribution that has a variance of 11.3 minutes², carry out the test for Brandon at 10% level of significance. Give your conclusion in the context of the question. [3]
- (iii) Explain what you understand by the phrase “at 10% level of significance” in the context of the question. [1]

- (iv) The same sample of 8 flights is now used to carry out a test at the 10% significance level, to test whether the mean flight times is different from what the airline claims. Using the answer in (ii), state the conclusion of this test. [2]

Another flight route travels from Portland to Los Angeles. Due to bad weather conditions, the flights sometimes take a longer time to arrive at Los Angeles. The airline claims that the mean flight time is k minutes. The operation manager of the airline believes that this number is understated and wants to test this claim. Sixty flights were randomly selected, and the mean flight time of 181 minutes and standard deviation of 13.6 minutes were recorded.

Find the range of values of k for which the claim by the airline is supported at 4% level of significance. Give your answer to the nearest minute. [4]

6. **NYJC Prelim 9758/2021/02/Q10**

An office worker, Natalie, has diabetes and has to monitor her blood glucose levels, which vary throughout the day. The results from a sample of 75 readings, x (in mmol/L), taken at random times over a week, are summarized by

$$\sum x = 511.5 \text{ and } \sum x^2 = 4027.89 .$$

- (i) Calculate unbiased estimates of the population mean and variance for the blood glucose levels. [2]
- (ii) Test at 5% significance level whether Natalie's mean blood glucose level, μ (in mmol/L), is greater than 6.0. You should state your hypotheses and give your conclusion in context. [4]
- (iii) State, giving a reason, whether the conclusion of the test in part (ii) would be valid if the 75 readings were all taken at weekends. [1]

Following a change in her diet, Natalie claims that her mean blood glucose level μ is now less than 6.0. She takes another random sample of 75 readings and notes that the total blood glucose levels is now 420. Using this sample, Natalie concludes that there is no reason to reject her claim at 6% level of significance.

- (iv) Find the range of possible values of the variance used in calculating the test statistic. [4]
- (v) Explain why there is no need for Natalie to know anything about the population distribution of the glucose blood levels when carrying out the tests in (ii) and (iv). [1]

7. **NJC Prelim 9758/2021/02/Q7**

The medical director of a hospital knows that the mean systolic blood pressure of patients who suffer from high blood pressure is 140 mmHg. He wishes to carry out a clinical trial to evaluate whether a new drug is effective in reducing the systolic blood pressure of patients who suffer from high blood pressure. The systolic blood pressure, x mmHg, of a random sample of 60 patients are summarized as follows.

$$\sum(x-140) = -37.6 \quad \sum(x-140)^2 = 1012.17$$

- (i) Calculate unbiased estimates of the population mean and variance of the systolic blood pressure of patients who suffer from high blood pressure [2]
- (ii) Carry out the test, at 5% level of significance, for the medical director. You should state your hypotheses and define any symbols that you use. [5]
- (iii) Upon closer inspection of the data of the sample of 60 patients, the director noted that the value of $\sum(x-140)$ is correct but the value of $\sum(x-140)^2$ should be larger instead. If a new test is carried out using this information at the 5% level of significance, explain whether the result of this test will differ from the result of the test in part (ii). [3]

8. **JPJC Prelim 9758/2021/02/Q10**

The management of a theme park claims that a visitor spends more than 9 hours in the theme park. The time, x hours, spent by 100 randomly chosen visitors is recorded and realized as follows:

$$\sum x = 920, \quad \sum x^2 = 8668.$$

- (i) Test at the 10% significance level whether the management's claim is valid. [6]
- (ii) Explain why the use of Central Limit Theorem is necessary in your calculation in (i). [1]

The management now claims that a visitor spends more than μ_0 hours in the theme park. A second random sample of 60 visitors is taken. The time, y hours, spent are recorded and the results are realized as follows:

$$\sum y = 550, \quad \sum y^2 = 5120.$$

- (iii) Combining the two samples into a single sample and using an algebraic method, calculate the set of values of μ_0 for the management's new claim not to be accepted at the 10% level of significance. (Answers obtained by trial and improvement from a calculator will obtain no marks.) [5]

9.

2017/RVHS/Prelim/II/Q11

- (i) ABC Electronic Company invested a huge sum of money in a new process in the production of a semiconductor device. The lifespan, X in hours, of this semiconductor device has mean 720 and variance 225. The production manager claims that the new process has increased the mean lifespan of the device. A random sample of 50 devices produced under the new process is examined and the sample mean lifespan in hours is denoted by $\bar{x}$. Find the range of values of $\bar{x}$ correct to 1 decimal place, if the manager's claim is rejected at the 5% level of significance. [4]
- (ii) It is later realized that the variance of the lifespan of the semiconductor device is wrongly quoted in part (i); and the production manager would like to conduct the test at 2.5% level of significance instead. Based on another sample of 50 devices, the following results are obtained:

$$\sum x = 36100, \sum x^2 = 26066890.$$

Find the p -value of the test and state the meaning of this p -value in the context of the question. Test whether the new process has increased the mean lifespan of the semiconductor device. [6]

- (iii) Using the p -value obtained in part (ii), state the conclusion if the manager decides to test at the 5% of level of significance whether the new production process has affected the mean lifespan of the semiconductor device. [2]

10.

2017/AJC/BT2/II/Q1

The manufacturer of metal rods claims that the average length of rods produced by his factory is 30 metres. A random sample of 60 metal rods is taken and the length, x metres, of each rod is measured. The results are summarized as follows:

$$\sum (x - 8) = 1400, \quad \sum (x - 8)^2 = 34800.$$

- (i) Find the unbiased estimates of the population mean and variance. [2]
- (ii) There was suspicion that the manufacturer has understated the mean length of the rods produced. Test at the 2% significance level whether this suspicion is true. [4]
- (iii) Suppose the null hypothesis is rejected in favour of the alternative hypothesis at α % level of significance. Determine the range of values of α . [1]
- (iv) Explain, in the context of the question, the meaning of "at the α % level of significance". [1]

11. 2017/IJC/BT2/I/Q11

Darry Ollen, a professional runner, claims that his average 100-meter sprint time is not more than 10 seconds. His rival, Barrick, disagrees with the claim and decides to conduct a research based on Darry's past records. A random sample of 50 races is taken and the time, x seconds, Darry took to complete a 100-meter sprint for each race is noted. The results are summarized by

$$\sum x = 522.35 \text{ and } \sum x^2 = 5586.31.$$

Test, at the 5% significance level, whether Darry's claim is valid. [8]

Darry undergoes intensive training sessions for the next few months, and the population standard deviation is given to be 1.5 seconds. Another random sample of 50 races is taken and the average time recorded is t seconds. A test at the 10% level of significance indicates that Darry's claim is now valid. Find the largest possible value of t , giving your answer correct to 2 decimal places. [3]

12. 2017/MI/BT2/II/Q11(modified)

The bakery manager claims that a bag of cookies weighs 250 g. An inspector wants to investigate whether the mean mass of a bag of cookies is overestimated. During the quality check on a random sample of 60 bags of cookies, the inspector finds that the mean mass is 249.3 g and the variance is 10 g^2 .

- (i) Carry out an appropriate test to determine whether the mean mass of a bag of cookies is overestimated at 8% level of significance. State, giving a reason, whether any assumption about the population is needed in order for the test to be valid. [6]
- (ii) Explain the meaning of 'at 8% level of significance' in the context of the question. [1]
- (iii) The inspector concludes that the sample could not provide sufficient evidence that the mean mass of a bag of cookies is overestimated at $\alpha\%$ significance level. State the range of values of α for him to draw such a conclusion. [1]
- (iv) The inspector wants to find out if the mean mass differs from 250 g at 8% level of significance. Using the same sample and result in part (i), explain whether there is a change in the conclusion. [2]

13. 2017/MJC/Prelim/II/Q8

In an assembly line, a machine is programmed to dispense shampoo into empty bottles and the volume of shampoo dispensed into each bottle is a normally distributed continuous random variable X . Under ordinary conditions, the expected value of X is 325 ml.

- (i) After a routine servicing of the machine, the assembly manager suspects that the machine is dispensing more shampoo than expected. A random sample of 60 bottles is taken and the data is as follows:

Volume of shampoo (correct to nearest ml)	324	325	326	327	328	329	330
Number of bottles	16	20	9	8	4	1	2

Find unbiased estimates of the population mean and variance, giving your answers to 2 decimal places. [2]

Test, at the 5% significance level, whether the assembly manager's suspicion is valid. [4]

Explain what it meant by the phrase 'at 5% significance level' in the context of the question. [1]

- (ii) Due to the assembly manager's suspicion, the machine is being recalibrated to dispense 325 ml of shampoo. Another random sample of 50 is taken and a two-tailed test, at the 5% significance level, concluded that the recalibration is done accurately. Given that the volume of shampoo dispensed into each bottle is normally distributed with standard deviation 1.2 ml, find the set of values the mean volume of the 50 bottles can take, giving your answers to 2 decimal places. [4]

14. RI Prelim 9758/2021/02/Q6

The recruitment manager of the private car hire company, I-ber, claims that the mean weekly earnings of a full-time driver is \$980. The managing director suspects that the mean weekly earnings is less than \$980 and he instructs the recruitment manager to carry out a hypothesis test on a sample of drivers. It is given that the population standard deviation of the weekly earnings is \$88.

- (i) State suitable hypotheses for the test, defining any symbols that you use. [2]

The recruitment manager takes a random sample of 10 drivers. He finds that the weekly earnings in dollars, are as follows.

942 950 905 1003 883 987 924 920 913 968

- (ii) Find the mean weekly earnings of the sample of these 10 drivers. Carry out the test, at 5% level of significance, for the recruitment manager. Give your conclusion in context and state a necessary assumption for the test to be valid. [5]

- (iii) Find the smallest level of significance at which the test would result in rejection of the null hypothesis, giving your answer correct to 1 decimal place. [1]

15. SAJC Prelim 9758/2021/02/Q8

A company that sells potato chips in packets claims that the mean amount of sodium content per packet is 798 mg.

- (a) It is given that the standard deviation of sodium content in the packets of potato chips is 6.5 mg. Upon receiving complaints that the potato chips were too salty, the local food agency chooses a large random sample of n packets of potato chips and found that the mean sodium content is 799.5 mg.

Given that the local food agency concludes at the 5% level of significance that the company has understated the amount of sodium content, find the set of values of n . [4]

- (b) The company decided to try out a new healthier recipe that reduces the amount of sodium content per packet. The company collected data from a random sample of 50 packets and found that the mean is 796.3 mg and standard deviation is 6.2 mg.

- (i) Test at 5% level of significance whether the mean amount of sodium in a packet of potato chips has been reduced from the original claim, after the introduction of the new healthier recipe. [4]

- (ii) Explain, in the context of the question, the meaning of “at 5% level of significance” for the test in (b)(i). [1]

Answers

Q1 (i) 11.5 ; 24.8

Q2 (iii) 467 ; 64.1

Q3 (ii) 31.8075 ; 0.245

Q4 (c) $\{\bar{y} \in \mathbb{R} : 0 \leq \bar{y} \leq 498.5\}$

Q5 $k > 178$ (3 s.f.) or $k \geq 178$

Q6 (i) 6.82 ; 7.29 (iv) $s^2 > 4.97$

Q7 (i) 139 ; 16.8

Q8 (iii) $\mu_0 > 9.06$

Q9 (i) $0 < \bar{x} < 723.4$ (ii) 0.0282

Q10 (i) $\frac{94}{3}$; 36.2 (iii) $\alpha \geq 4.30$

Q11 10.27

Q12 (iii) $\alpha < 4.45$

Q13 (i) 325.58 ; 2.35 (ii) $324.67 < \bar{x} < 325.33$

Q14 (ii) $\bar{x} = 939.5$ (iii) 7.3%

Q15 (a) $\{n \in \mathbb{Z}^+ : n \geq 51\}$